

A Note on Cauchy's Limit Theorems and Foundational Limit Evaluation

Bypassing Stolz–Cesàro for Subjective Entrance Evaluations

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1 Core Theorems and Proofs

In rigorous subjective evaluations such as the ISI UGB, examiners reward arguments built from first principles. A proof that invokes the Stolz–Cesàro theorem without supplying its derivation often sidesteps precisely the algebraic or analytic content the problem is designed to test. Cauchy's limit theorems and the Squeeze Theorem are universally accepted alternatives that admit short, transparent ϵ – N proofs and dispatch the same family of problems with cleaner structural reasoning.

1.1 Cauchy's First Limit Theorem (Arithmetic Mean)

Theorem 1

If a sequence (x_n) converges to a finite limit L , then the sequence of its arithmetic means also converges to L :

$$\lim_{n \rightarrow \infty} x_n = L \implies \lim_{n \rightarrow \infty} \frac{x_1 + x_2 + \cdots + x_n}{n} = L.$$

Proof. Define $y_n = x_n - L$. Since $\lim_{n \rightarrow \infty} x_n = L$, it follows that $\lim_{n \rightarrow \infty} y_n = 0$. We must show that

$$\lim_{n \rightarrow \infty} \frac{y_1 + y_2 + \cdots + y_n}{n} = 0.$$

By the definition of a limit, for any given $\epsilon > 0$ there exists an integer $N_1 \in \mathbb{N}$ such that for all $n > N_1$,

$$|y_n| < \frac{\epsilon}{2}.$$

For $n > N_1$, split the arithmetic mean into the first N_1 terms and the remaining ones:

$$\left| \frac{1}{n} \sum_{i=1}^n y_i \right| \leq \frac{1}{n} \sum_{i=1}^{N_1} |y_i| + \frac{1}{n} \sum_{i=N_1+1}^n |y_i|.$$

Let $S = \sum_{i=1}^{N_1} |y_i|$. Since N_1 is fixed for the chosen ϵ , S is a constant real number. For the second sum, $|y_i| < \epsilon/2$ for all $i > N_1$, so

$$\frac{1}{n} \sum_{i=N_1+1}^n |y_i| < \frac{n - N_1}{n} \cdot \frac{\epsilon}{2} < \frac{\epsilon}{2}.$$

The inequality becomes

$$\left| \frac{1}{n} \sum_{i=1}^n y_i \right| < \frac{S}{n} + \frac{\epsilon}{2}.$$

Because S is constant, $\lim_{n \rightarrow \infty} S/n = 0$, so there exists an integer $N_2 \in \mathbb{N}$ such that for all $n > N_2$,

$$\frac{S}{n} < \frac{\epsilon}{2}.$$

Choose $N = \max(N_1, N_2)$. For all $n > N$, both bounds hold:

$$\left| \frac{1}{n} \sum_{i=1}^n y_i \right| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

The arithmetic mean of y_n converges to 0, which gives the convergence of the arithmetic mean of x_n to L . ■

1.2 Cauchy's Second Limit Theorem (Geometric Mean)

Theorem 2

If (x_n) is a sequence of strictly positive real numbers and $\lim_{n \rightarrow \infty} x_n = L$, then the sequence of its geometric means also converges to L :

$$\lim_{n \rightarrow \infty} (x_1 \cdot x_2 \cdots x_n)^{1/n} = L.$$

Proof. We treat the cases $L > 0$ and $L = 0$ separately.

Case 1: $L > 0$. Let $z_n = \ln(x_n)$. Since the natural logarithm is continuous on the strictly positive reals,

$$\lim_{n \rightarrow \infty} z_n = \ln\left(\lim_{n \rightarrow \infty} x_n\right) = \ln(L).$$

Applying Cauchy's First Limit Theorem to (z_n) ,

$$\lim_{n \rightarrow \infty} \frac{\ln(x_1) + \ln(x_2) + \cdots + \ln(x_n)}{n} = \ln(L),$$

that is,

$$\lim_{n \rightarrow \infty} \ln\left((x_1 \cdot x_2 \cdots x_n)^{1/n}\right) = \ln(L).$$

Exponentiating both sides, by continuity of e^x ,

$$\lim_{n \rightarrow \infty} (x_1 \cdot x_2 \cdots x_n)^{1/n} = e^{\ln(L)} = L.$$

Case 2: $L = 0$. Since $x_n \rightarrow 0$ and $x_n > 0$, for any $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $0 < x_n < \epsilon$ for all $n > N$. For $n > N$, split the geometric mean:

$$(x_1 \cdot x_2 \cdots x_n)^{1/n} = (x_1 \cdots x_N)^{1/n} \cdot (x_{N+1} \cdots x_n)^{1/n}.$$

Bounding the second factor,

$$(x_{N+1} \cdots x_n)^{1/n} < (\epsilon^{n-N})^{1/n} = \epsilon^{1-N/n}.$$

Hence

$$0 < (x_1 \cdot x_2 \cdots x_n)^{1/n} < (x_1 \cdots x_N)^{1/n} \cdot \epsilon^{1-N/n}.$$

As $n \rightarrow \infty$, the constant block $(x_1 \cdots x_N)^{1/n} \rightarrow 1$ and $\epsilon^{1-N/n} \rightarrow \epsilon$. For sufficiently large n , the geometric mean lies strictly between 0 and a value arbitrarily close to ϵ , so the limit is 0. ■

Corollary 2.1 (The Ratio-to-Root Limit)

If (a_n) is a sequence of positive real numbers and $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L$, then $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = L$.

Proof. Construct (x_n) by setting $x_1 = a_1$ and $x_n = a_n/a_{n-1}$ for $n \geq 2$. By condition, $\lim_{n \rightarrow \infty} x_n = L$. The product of the first n terms telescopes:

$$x_1 \cdot x_2 \cdots x_n = a_1 \cdot \frac{a_2}{a_1} \cdots \frac{a_n}{a_{n-1}} = a_n.$$

By Cauchy's Second Limit Theorem,

$$\lim_{n \rightarrow \infty} (x_1 \cdot x_2 \cdots x_n)^{1/n} = \lim_{n \rightarrow \infty} x_n = L \implies \lim_{n \rightarrow \infty} (a_n)^{1/n} = L. \quad \blacksquare$$

2 The Squeeze (Sandwich) Theorem

When a limit involves expanding sums, products, or denominators that do not match the structure of an arithmetic or geometric mean, the cleanest analytic tool is to construct sharp inequalities and squeeze the target sequence between them.

Theorem 3

Let (a_n) , (b_n) , and (c_n) be real sequences. If there exists an integer $N_0 \in \mathbb{N}$ such that for all $n \geq N_0$,

$$a_n \leq b_n \leq c_n,$$

and $\lim_{n \rightarrow \infty} a_n = L = \lim_{n \rightarrow \infty} c_n$, then (b_n) converges and $\lim_{n \rightarrow \infty} b_n = L$.

Proof. For any given $\epsilon > 0$, since $\lim_{n \rightarrow \infty} a_n = L$, there exists $N_1 \in \mathbb{N}$ such that for all $n > N_1$,

$$|a_n - L| < \epsilon \implies L - \epsilon < a_n < L + \epsilon.$$

Similarly, since $\lim_{n \rightarrow \infty} c_n = L$, for the same ϵ there exists $N_2 \in \mathbb{N}$ such that for all $n > N_2$,

$$|c_n - L| < \epsilon \implies L - \epsilon < c_n < L + \epsilon.$$

Set $N = \max(N_0, N_1, N_2)$. For all $n > N$, the bounding condition and both limit conditions hold simultaneously, so

$$L - \epsilon < a_n \leq b_n \leq c_n < L + \epsilon,$$

and therefore $L - \epsilon < b_n < L + \epsilon$, i.e. $|b_n - L| < \epsilon$. Since $\epsilon > 0$ was arbitrary, $\lim_{n \rightarrow \infty} b_n = L$. $\blacksquare$

3 Application Problems (Bypassing Stolz–Cesàro)

The following problems invite long expanding-difference computations under Stolz–Cesàro but yield cleanly to Cauchy's theorems or to the Squeeze Theorem.

Problem 1

Evaluate

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sqrt[n]{(n+1)(n+2) \cdots (n+n)}.$$

Solution. Pull the factor $1/n$ inside the n -th root and set

$$a_n = \frac{(n+1)(n+2) \cdots (2n)}{n^n}.$$

By the Ratio-to-Root corollary,

$$\frac{a_{n+1}}{a_n} = \frac{(2n+1)(2n+2)}{(n+1)^2} \cdot \left(\frac{n}{n+1}\right)^n = \frac{2(2n+1)}{n+1} \cdot \frac{1}{\left(1+\frac{1}{n}\right)^n}.$$

As $n \rightarrow \infty$, this tends to $4 \cdot \frac{1}{e} = \frac{4}{e}$. Hence $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \frac{4}{e}$. ■

Problem 2

Evaluate

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(\frac{k}{k+1}\right)^k.$$

Solution. This is the arithmetic mean of $y_k = \left(\frac{k}{k+1}\right)^k$. Compute the limit of the general term:

$$\lim_{k \rightarrow \infty} y_k = \lim_{k \rightarrow \infty} \frac{1}{\left(1+\frac{1}{k}\right)^k} = \frac{1}{e}.$$

By Cauchy's First Limit Theorem, the arithmetic mean shares this limit. The result is $\frac{1}{e}$. ■

Problem 3

Evaluate

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{\sqrt{n^4 + k}}.$$

Solution. The Squeeze Theorem bypasses Stolz–Cesàro and Riemann-sum tricks here. Write S_n for the sum and bound the terms by their values at $k = n$ (smallest summand) and $k = 1$ (largest summand):

$$\frac{1}{\sqrt{1 + \frac{1}{n^3}}} \leq S_n \leq \frac{1}{\sqrt{1 + \frac{1}{n^4}}}.$$

As $n \rightarrow \infty$, both bounds converge to 1. By the Squeeze Theorem, $\lim_{n \rightarrow \infty} S_n = 1$. ■

Problem 4

Evaluate

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n k \sin\left(\frac{1}{k}\right).$$

Solution. The expression is the arithmetic mean of $x_k = k \sin(1/k)$. The general-term limit is

$$\lim_{k \rightarrow \infty} k \sin\left(\frac{1}{k}\right) = \lim_{k \rightarrow \infty} \frac{\sin(1/k)}{1/k} = 1.$$

By Cauchy's First Limit Theorem, the arithmetic mean shares this limit. The result is 1. ■

Problem 5

Evaluate

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt[n]{n!}}.$$

Solution. Rewrite as $\lim_{n \rightarrow \infty} \sqrt[n]{n^n/n!}$ and set $a_n = n^n/n!$. By the Ratio-to-Root theorem,

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1}}{(n+1)!} \cdot \frac{n!}{n^n} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e.$$

The root limit therefore equals e . ■

Problem 6

Evaluate

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(1 + \frac{1}{k}\right)^k.$$

Solution. The expression is the arithmetic mean of $y_k = \left(1 + \frac{1}{k}\right)^k$. Since the underlying sequence satisfies $\lim_{k \rightarrow \infty} y_k = e$, Cauchy's First Limit Theorem gives an arithmetic-mean limit of exactly e . ■

Problem 7

Evaluate

$$\lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}}{\ln n}.$$

Solution. Write $H_n = \sum_{k=1}^n \frac{1}{k}$ for the n -th harmonic number; the task is to evaluate $\lim_{n \rightarrow \infty} H_n / \ln n$.

Construct rigorous upper and lower bounds for H_n by comparing the discrete sum to the integral of the strictly decreasing function $f(x) = 1/x$. For any integer $k \geq 2$, the area under the curve over $[k, k+1]$ is bounded above by the left-endpoint rectangle of height $1/k$ and width 1, and the area over $[k-1, k]$ is bounded below by the right-endpoint rectangle of the same height:

$$\int_k^{k+1} \frac{1}{x} dx < \frac{1}{k} < \int_{k-1}^k \frac{1}{x} dx.$$

Summing from $k = 2$ to n ,

$$\int_2^{n+1} \frac{1}{x} dx < H_n - 1 < \int_1^n \frac{1}{x} dx,$$

that is,

$$\ln(n+1) - \ln 2 < H_n - 1 < \ln n.$$

Adding 1 to all terms isolates H_n :

$$\ln(n+1) - \ln 2 + 1 < H_n < \ln n + 1.$$

Dividing through by $\ln n$ (which is strictly positive for $n > 1$),

$$\frac{\ln(n+1) - \ln 2 + 1}{\ln n} < \frac{H_n}{\ln n} < \frac{\ln n + 1}{\ln n}.$$

Set $a_n = \frac{\ln(n+1) - \ln 2 + 1}{\ln n}$ and $c_n = \frac{\ln n + 1}{\ln n}$.

For the lower bound,

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(\frac{\ln(n(1 + \frac{1}{n}))}{\ln n} - \frac{\ln 2 - 1}{\ln n} \right) = \lim_{n \rightarrow \infty} \left(1 + \frac{\ln(1 + \frac{1}{n})}{\ln n} \right) = 1 + 0 = 1.$$

For the upper bound,

$$\lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{\ln n} \right) = 1.$$

Both bounds converge to 1, so the Squeeze Theorem applies and

$$\lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}}{\ln n} = 1. \quad \blacksquare$$